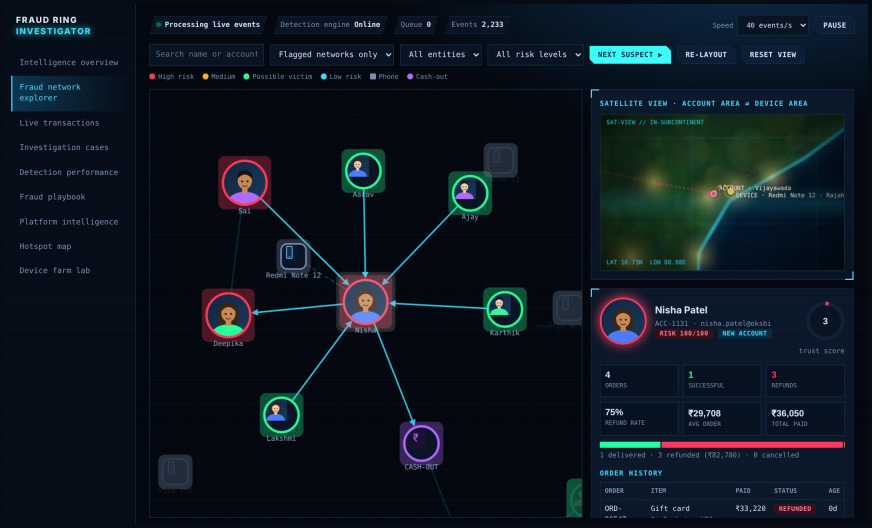
FRAUD DETECTION · JAVA

Fraud Ring Investigator

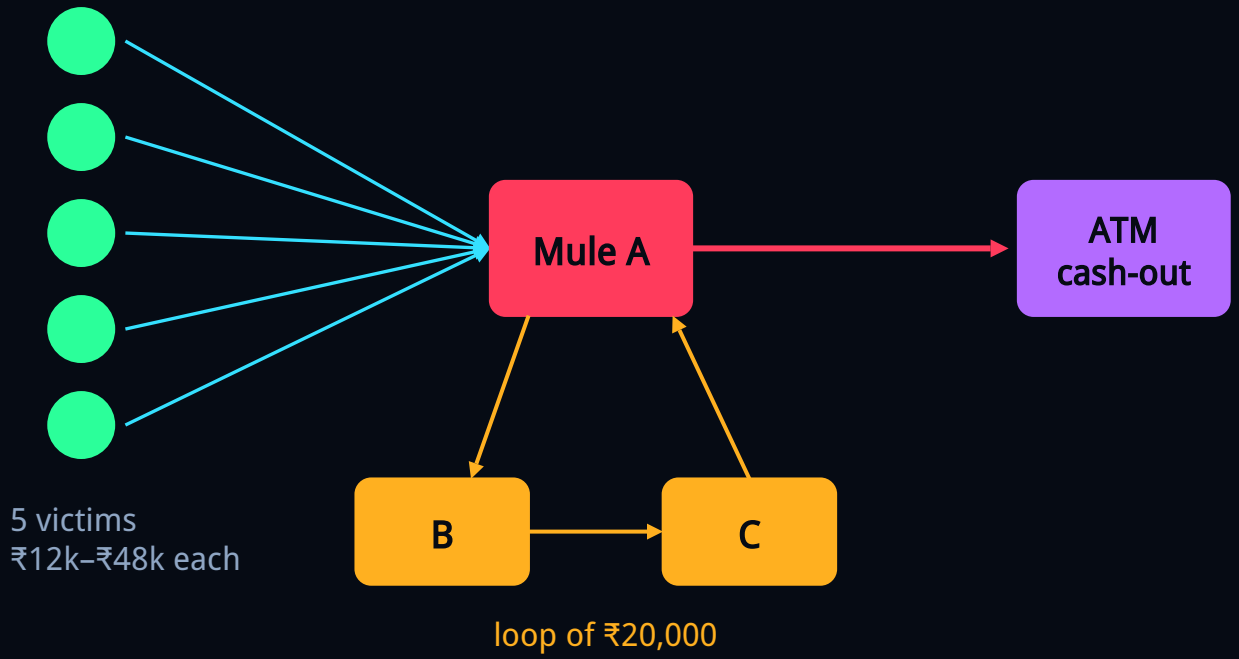
Catch mule rings and device farms in seconds, not weeks



Zero dependencies · 26 tests · synthetic data

Presented by [YOUR NAME]

Every payment looks normal



≈ 10 s

from first payment to the money leaving

Each hop is small, so a rule like “flag above ₹50,000” sees nothing.

Judge the network, not the payment

ONE PAYMENT AT A TIME

Rule: flag anything above ₹50,000

Every hop is under the limit.

Nothing flagged

VS

THE WHOLE NETWORK

Shared phone + money loop + fast cash-out

Links accounts, phones and time.

Risk 100 · HIGH

Four patterns, one score

+35 — **Shared device**
3+ accounts, 2+ different surnames, one phone

+25 — **Rapid cash-out**
80% of a ₹50,000+ inflow leaves in seconds

+25 — **Circular transfers**
Money returns to its origin in 3–4 hops (DFS)

+15 — **Fan-in**
4+ people pay one account, ₹50,000+, in a minute

Every score is explained

CASE-002 risk 100 HIGH
+35 Shared device: 3 unrelated accounts, one phone
+25 Rapid cash-out: INR 144,892 in, INR 140,546
    out within seconds
+25 Circular: Lakshmi -> Fatima -> Deepika -> Lakshmi
+15 Fan-in: 4 people paid one account
    INR 99,139 within a minute

Real output from the engine: demo --seed 7

HIGH	70 and above
MEDIUM	40 to 69
LOW	under 40

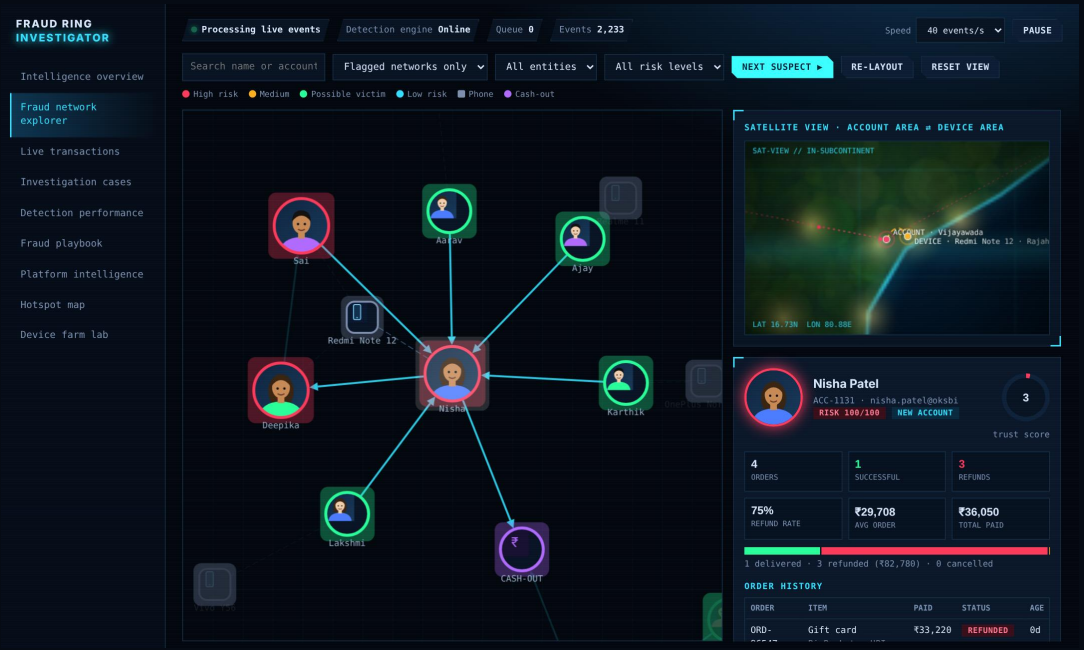
Alerts on the same account or phone merge into one case, with notes, status and a full audit trail.

A live investigation dashboard

1. **Click any node**
   profile, orders, refunds, risk

2. **Colour tells the story**
   red suspect, green victim, square = phone

3. **Satellite view**
   account area vs device area



One phone, many accounts

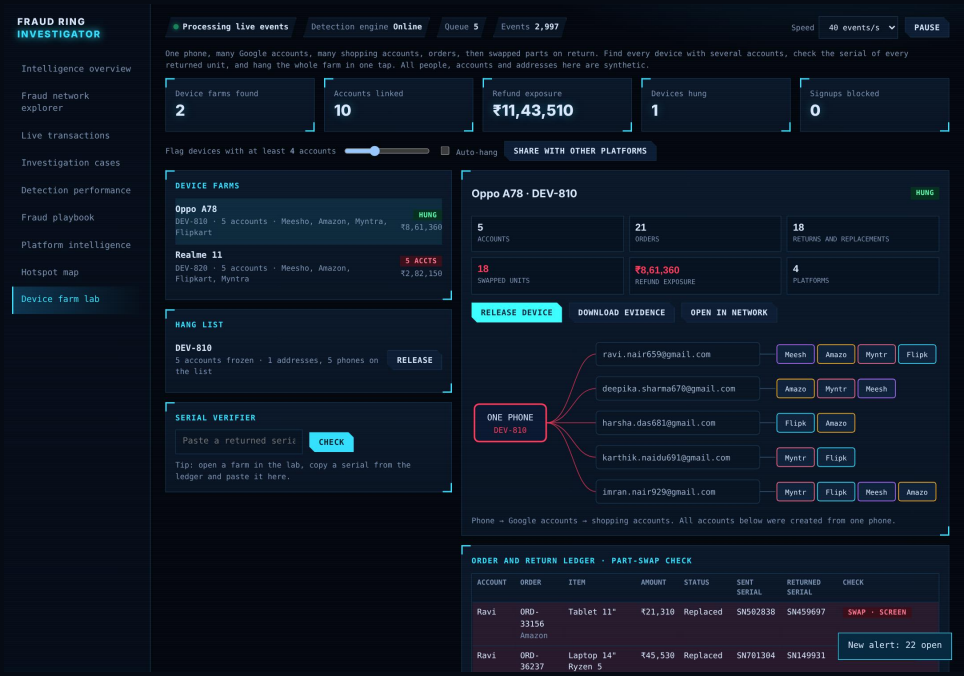
findFarms
phones with many accounts

verifySerial
sent vs returned serial

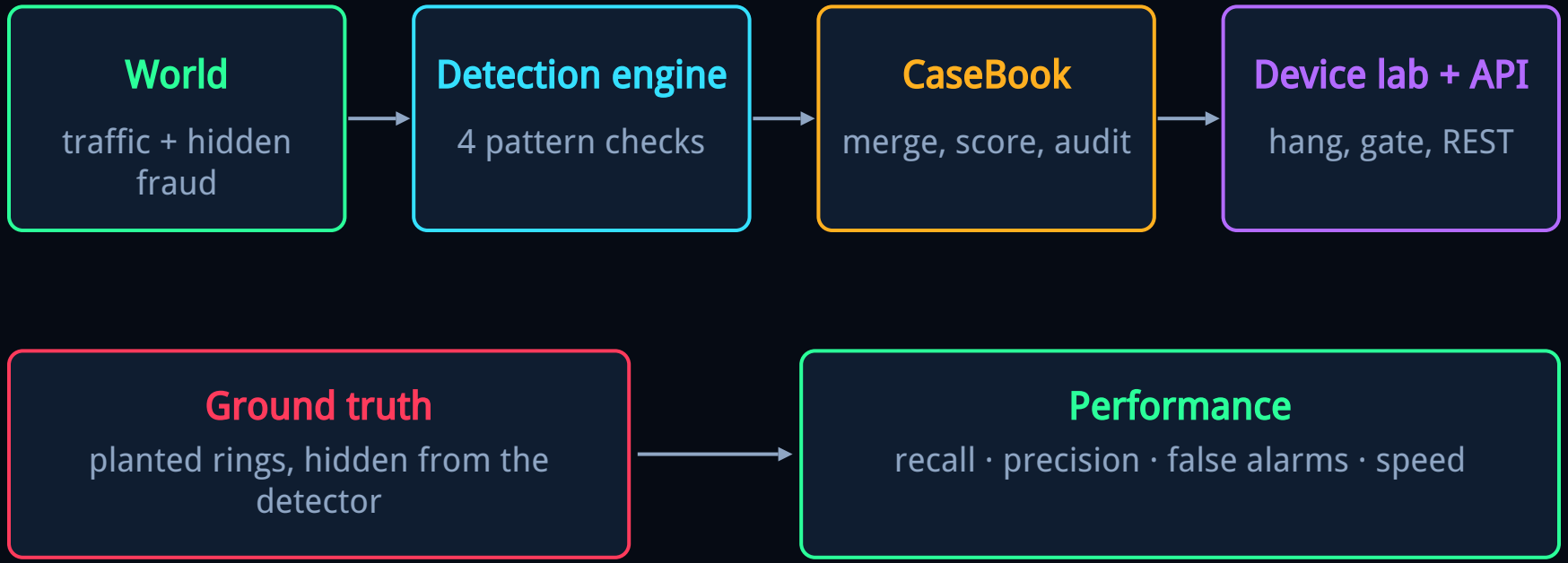
hang / release
freeze a whole farm at once

gate
ALLOWED · REVIEW · BLOCKED

evidenceReport
ready for trust & safety

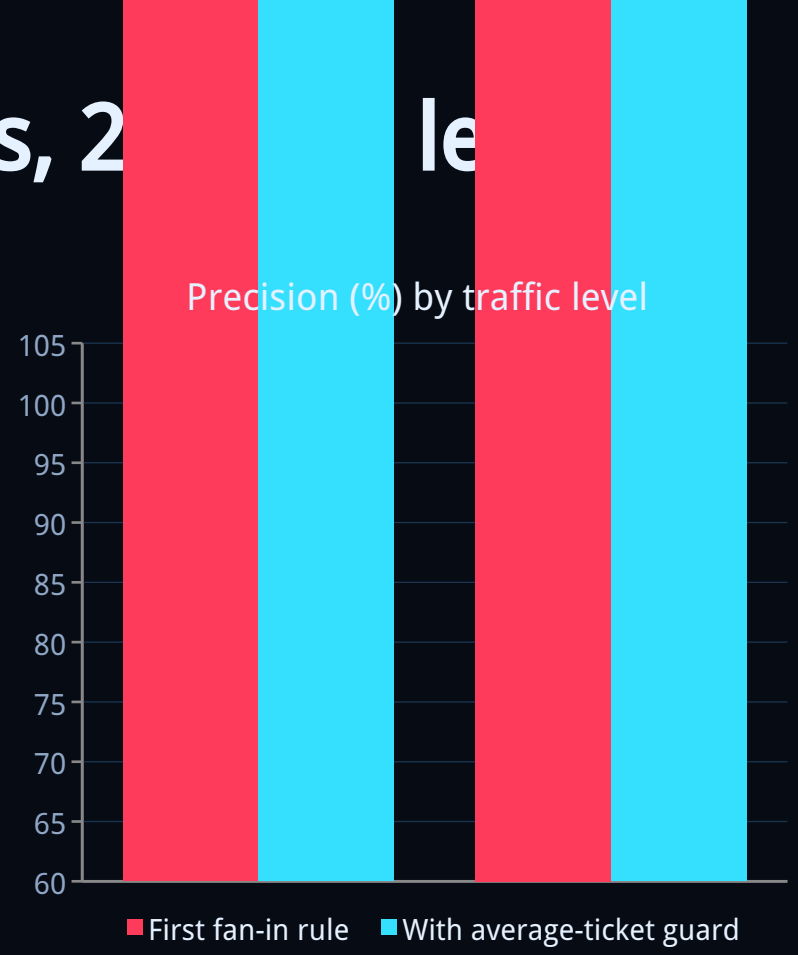


Simulate, detect, act



Results: 100 seeds, 2 le

1,629 / 1,629 fraud groups caught	**0** false alarms
3.8 s mean time to flag	**26 / 26** tests passing



What went wrong, and the fix

1 Busy-minute false alarms

Precision fell 100% → 85% at 8× traffic

Fix: minimum average payment ₹8,000

2 A “miss” that was not

Recall 94%: ring planted 1 s before the end

Fix: 30-second grace window

3 A test that lied

API returned bad JSON; the test still passed

Fix: assert exact JSON keys

4 Dashboard said 50% precision

Planted farms scored as false alarms

Fix: count farms as ground truth

What shaped the design

1	**Mule accounts and UPI loops**	→	fan-in, rapid cash-out, circular transfers
2	**Phone farms and return item-swaps**	→	device lab, serial verifier, hang and gate
3	**Families and shoppers that look odd**	→	false-positive guards and a zero-fraud control run

Public sources: cybercrime.gov.in and the 1930 helpline, RBI and NPCI fraud awareness material, graph-analytics basics. General patterns only, never accusations.

Five case studies

1

Mule ring

All four patterns fire in about 4 s

2

Device farm

Five accounts, swapped serials, ₹787k exposure

3

Family phone

Correctly not flagged

4

Busy minute

A false alarm we found and fixed

5

Slow ring

A known miss, documented

Each one has the output, the investigator's next steps and what could fool it: docs/case-studies/

What it cannot do (yet)

Synthetic data only

Real fraud is messier and adapts

Self-confirming benchmark

The simulator plants what the detector hunts

Hand-set thresholds

₹10,000 · 4 senders · 80% · ₹8,000 need tuning

Evasion works

Slow, split rings with one phone per mule are missed

A score is a lead, not proof. A human reviews before any account is frozen.

NEXT

Make it real

- Labelled, privacy-safe data
- Learned and adaptive thresholds
- Graph machine learning next to the rules
- Analyst feedback loop and a database

Open source · MIT

github.com/YOUR-USERNAME/fraud-ring-investigator

Live demo on GitHub Pages.
Thank you!